MBI802 DATABASE MANAGEMENT SYSTEMS

ER Diagram Tutorial: Answers

Model answers for Tasks 1 to 6

Lecturer **Yasas Sri Wickramasinghe** Use with **ER Diagram Tutorial**

How to use these answers

1. Try each task on your own before you look at its answer.
2. Check one part at a time: entities first, then attributes, then relationships, then the diagram.
3. If a part is different from yours, read the reason given, fix your diagram and continue.

Your layout can be different from the answer. Your diagram is correct if it has the same entities, keys, relationships and cardinalities.

Contents

Task 1	Customers and accounts (simple)
Task 2	Students and clubs (simple)
Task 3	Wards, nurses, doctors and patients (moderate)
Task 4	Branches, employees and joint accounts (moderate)
Task 5	Courses, lecturers and assignments (moderate)
Task 6	Customers and purchases (simple)

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TASK 1 Simple · Bank · About 10 minutes · Answers

Customers and accounts

1. Entities

Entity	Reason
`CUSTOMER`	The bank stores details about many customers.
`ACCOUNT`	The bank stores details about many accounts.

Not entities: The bank is the whole system, so it is not drawn as an entity. Mere uses the database, so she is not an entity either.

2. Attributes

Entity	Key attribute	Other attributes
`CUSTOMER`	`CustomerID`	`FirstName, LastName, PhoneNumber`
`ACCOUNT`	`AccountNumber`	`AccountType, Balance`

Two customers can have the same name, but not the same CustomerID. That is why CustomerID is the key.

3. Relationships

Relationship	The two questions		Cardinality
`CUSTOMER OWNS ACCOUNT`	One CUSTOMER can own how many ACCOUNTs? One ACCOUNT is owned by how many CUSTOMERs?	**Many** **One**	**1:N** Write 1 next to CUSTOMER and N next to ACCOUNT.

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4. ER diagram

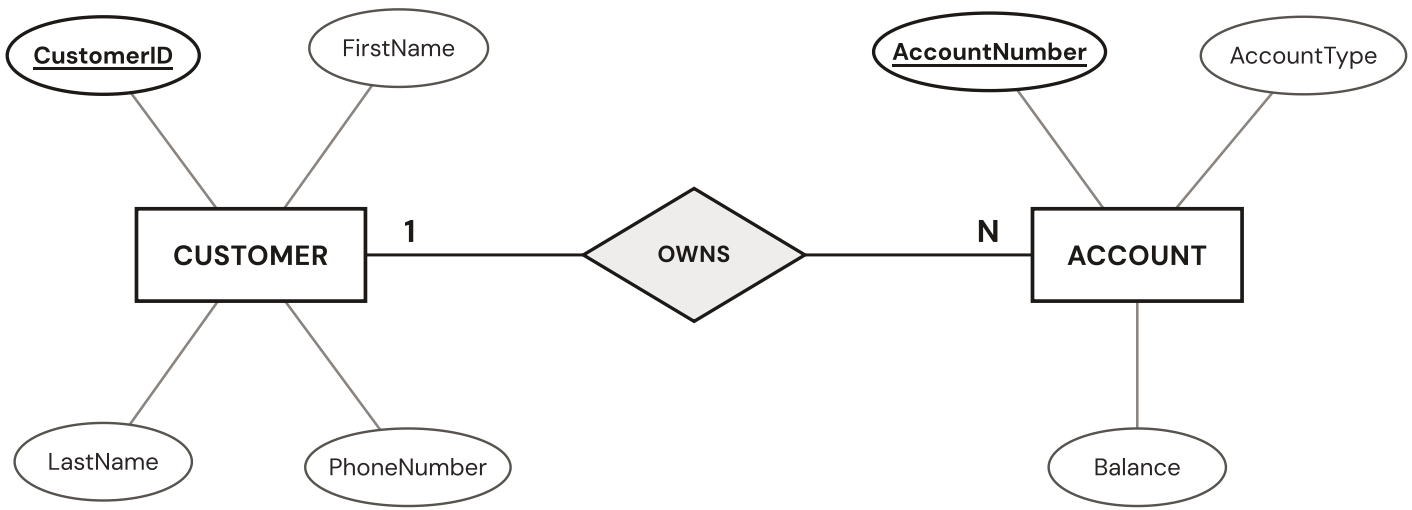


Figure 1. ER diagram for Task 1, customers and accounts.

Reading the diagram

- One CUSTOMER OWNS many ACCOUNTs.
- Each ACCOUNT is owned by one CUSTOMER.

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TASK 2 Simple · University · About 10 minutes · Answers

Students and clubs

1. Entities

Entity	Reason
`STUDENT`	Sione stores details about many students.
`CLUB`	Sione stores details about many clubs.

Not entities: The university, the student centre and Sione are not entities. Chess, hiking and cooking are not entities either. They are examples of clubs, so they are values of ClubName.

2. Attributes

Entity	Key attribute	Other attributes
`STUDENT`	`StudentID`	`FirstName, LastName, Email`
`CLUB`	`ClubID`	`ClubName, MeetingDay`

DateJoined is not listed here because it belongs to the relationship, not to STUDENT or CLUB.

3. Relationships

Relationship	The two questions		Cardinality
`STUDENT JOINS CLUB` `Relationship attributes: DateJoined`	One STUDENT can join how many CLUBs? One CLUB can have how many STUDENTs?	**Many** **Many**	**M:N** Write M next to STUDENT and N next to CLUB.

STUDENT JOINS CLUB: Both answers are “many”, so the relationship is many-to-many. DateJoined is connected to the JOINS diamond.

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4. ER diagram

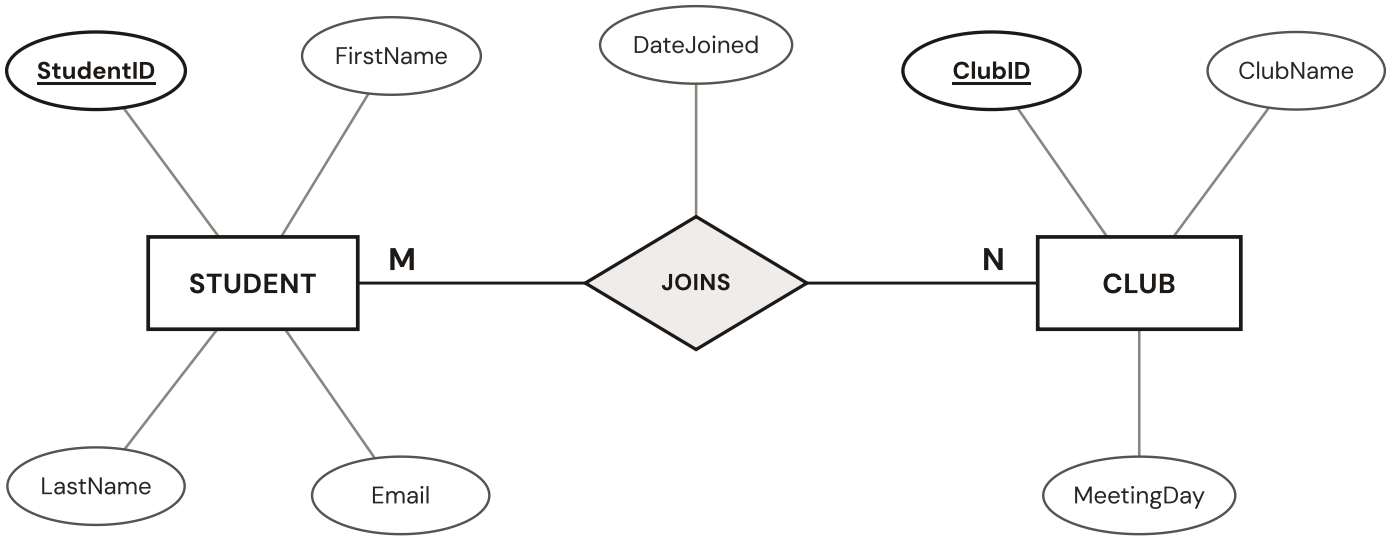


Figure 2. ER diagram for Task 2, students and clubs.

Reading the diagram

- A STUDENT JOINS many CLUBs.
- A CLUB has many STUDENTs.
- DateJoined records when a student joined a club.

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TASK 3 Moderate · Hospital · About 20 minutes · Answers

Wards, nurses, doctors and patients

1. Entities

Entity	Reason
`WARD`	The hospital stores details about each ward.
`PATIENT`	The hospital stores details about many patients.
`NURSE`	The hospital stores details about many nurses.
`DOCTOR`	The hospital stores details about many doctors.

Not entities: The hospital and the whiteboards are not entities. Priya is a nurse, so she would be stored as one row in NURSE. She does not need her own entity.

2. Attributes

Entity	Key attribute	Other attributes
`WARD`	`WardNumber`	`WardName, NumberOfBeds`
`PATIENT`	`PatientID`	`FirstName, LastName, DateOfBirth`
`NURSE`	`NurseID`	`FirstName, LastName, Shift`
`DOCTOR`	`DoctorID`	`FirstName, LastName, Specialty`

Store DateOfBirth, not Age. Age changes every year, but a date of birth does not. NURSE and DOCTOR both have FirstName and LastName. That is fine, because each attribute belongs to its own entity.

3. Relationships

Relationship	The two questions		Cardinality
`PATIENT STAYS_IN WARD`	One PATIENT stays in how many WARDs? One WARD has how many PATIENTs?	One Many	**N:1** Write N next to PATIENT and 1 next to WARD.
`NURSE WORKS_IN WARD`	One NURSE works in how many WARDs? One WARD has how many NURSEs?	One Many	**N:1** Write N next to NURSE and 1 next to WARD.
`DOCTOR TREATS PATIENT` `Relationship attributes: TreatmentDate, Diagnosis`	One DOCTOR treats how many PATIENTs? One PATIENT is treated by how many DOCTORs?	Many Many	**M:N** Write M next to DOCTOR and N next to PATIENT.

PATIENT STAYS_IN WARD: N:1 is the same as 1:N read from the other side: one ward, many patients.

DOCTOR TREATS PATIENT: TreatmentDate and Diagnosis describe one doctor treating one patient, so they are connected to TREATS.

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4. ER diagram

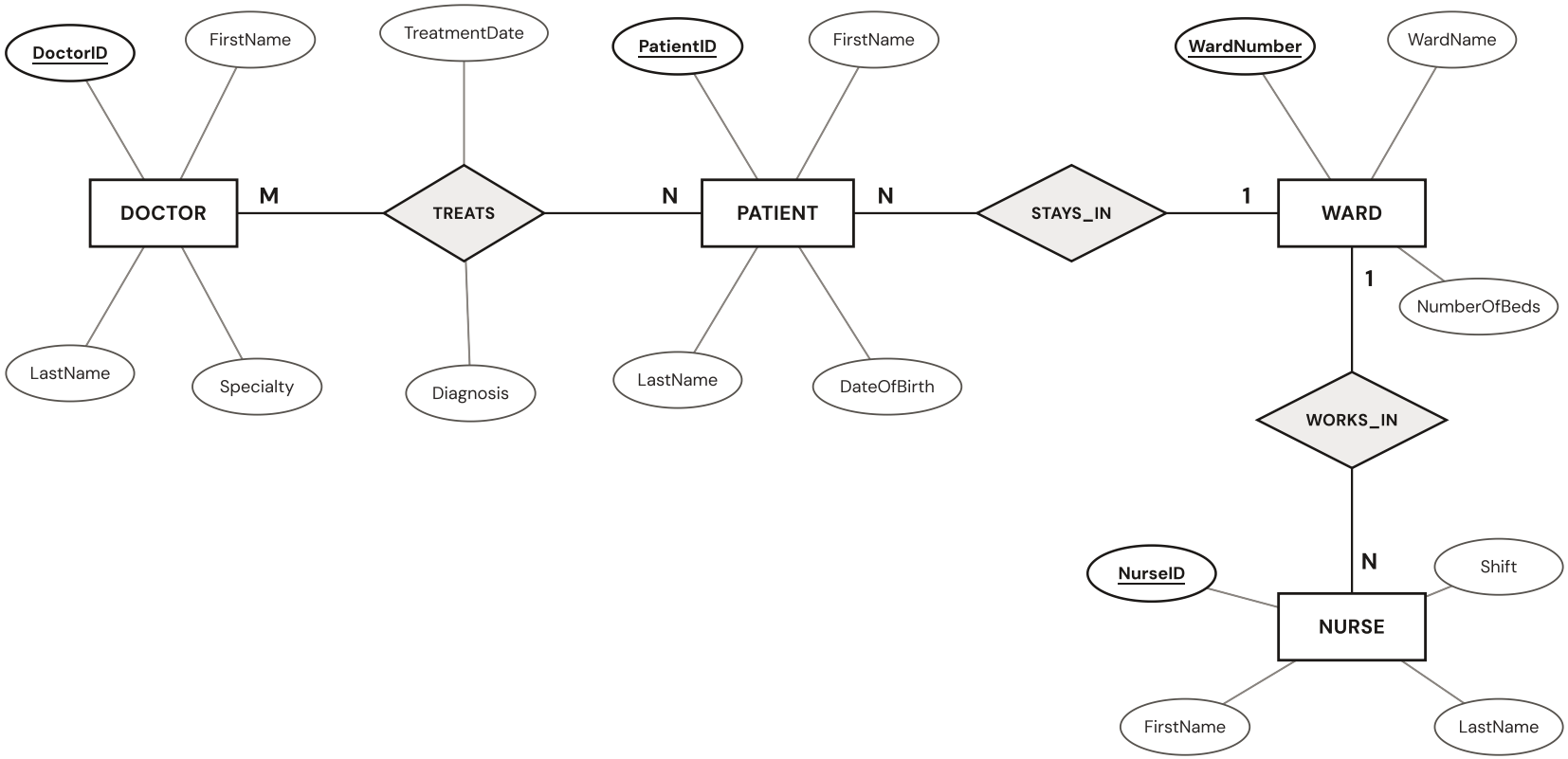


Figure 3. ER diagram for Task 3, wards, nurses, doctors and patients.

Reading the diagram

- A PATIENT STAYS_IN one WARD. A WARD has many PATIENTs.
- A NURSE WORKS_IN one WARD. A WARD has many NURSEs.
- A DOCTOR TREATS many PATIENTs. A PATIENT can be treated by many DOCTORs.
- Each treatment records a TreatmentDate and a Diagnosis.

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TASK 4 Moderate · Bank · About 25 minutes · Answers

Branches, employees and joint accounts

1. Entities

Entity	Reason
`CUSTOMER`	Same as Task 1.
`ACCOUNT`	Same as Task 1.
`BRANCH`	The bank stores details about each branch.
`EMPLOYEE`	The bank stores details about many employees.

Not entities: A joint account is still an ACCOUNT. "Joint" means it can have more than one owner, and that is shown by the cardinality. A manager is not a separate entity either. A manager is an EMPLOYEE who also MANAGES a branch.

2. Attributes

Entity	Key attribute	Other attributes
`CUSTOMER`	`CustomerID`	`FirstName, LastName, PhoneNumber`
`ACCOUNT`	`AccountNumber`	`AccountType, Balance`
`BRANCH`	`BranchCode`	`BranchName, City`
`EMPLOYEE`	`EmployeeID`	`FirstName, LastName, JobTitle`

DateAdded is not listed here. It describes one customer and one account together, so it belongs to the OWNS relationship.

3. Relationships

Relationship	The two questions		Cardinality
`CUSTOMER OWNS ACCOUNT` `Relationship attributes: DateAdded`	One CUSTOMER can own how many ACCOUNTs? One ACCOUNT can be owned by how many CUSTOMERs?	**Many** **Many (joint accounts)**	**M:N** Write M next to CUSTOMER and N next to ACCOUNT.
`BRANCH HOLDS ACCOUNT`	One BRANCH holds how many ACCOUNTs? One ACCOUNT is held at how many BRANCHes?	**Many** **One**	**1:N** Write 1 next to BRANCH and N next to ACCOUNT.
`EMPLOYEE WORKS_AT BRANCH`	One EMPLOYEE works at how many BRANCHes? One BRANCH has how many EMPLOYEEs?	**One** **Many**	**N:1** Write N next to EMPLOYEE and 1 next to BRANCH.
`EMPLOYEE MANAGES BRANCH`	One EMPLOYEE manages how many BRANCHes? One BRANCH is managed by how many EMPLOYEEs?	**One at most** **One**	**1:1** Write 1 on both sides.

CUSTOMER OWNS ACCOUNT: The entities are the same as in Task 1, but the rule has changed, so the cardinality has changed from 1:N to M:N. Always use the rules given in the scenario.

EMPLOYEE MANAGES BRANCH: Both answers are "one", so this relationship is one-to-one.

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TASK 4 Moderate · Bank · About 25 minutes · Answers

4. ER diagram

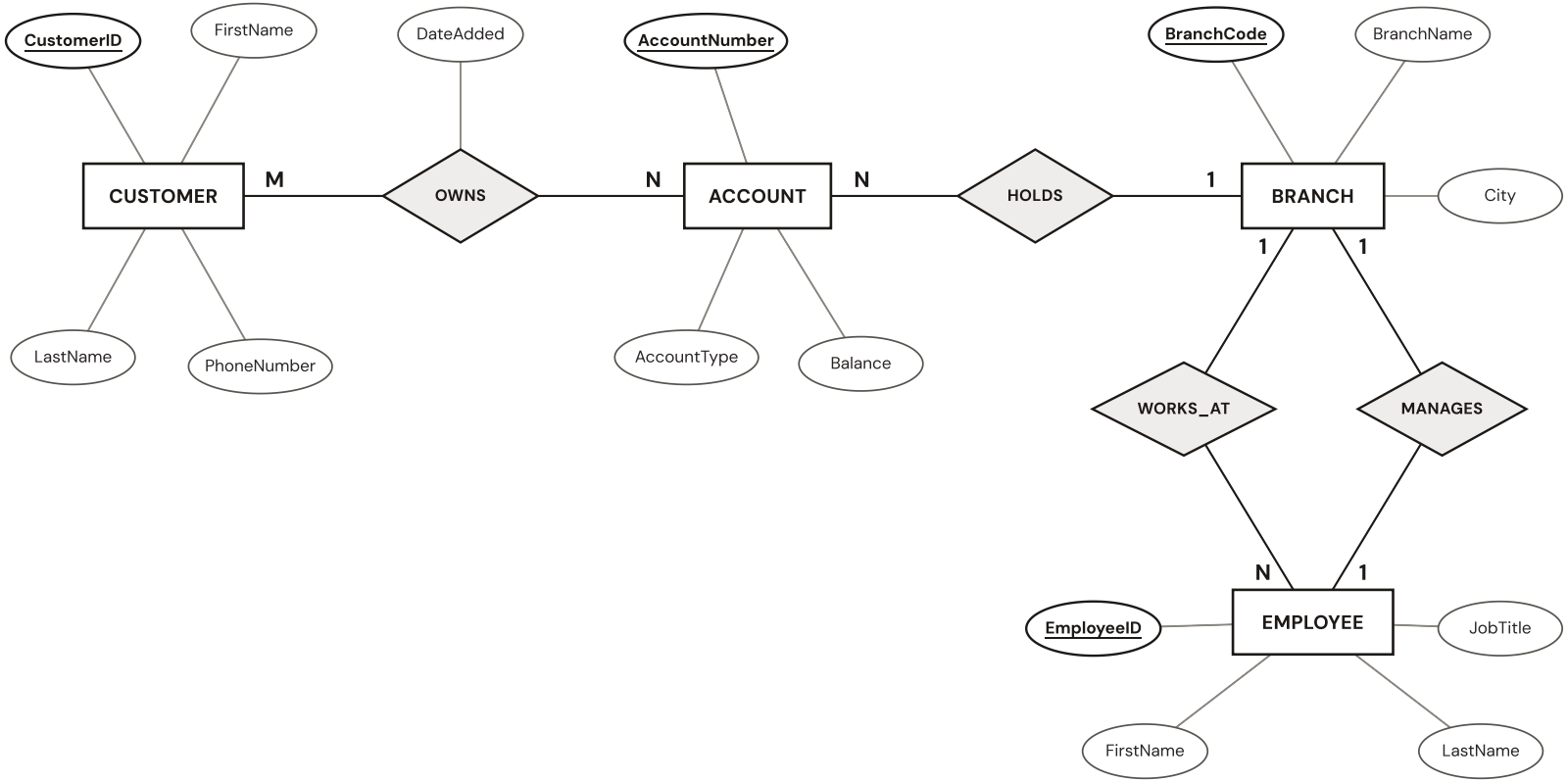


Figure 4. ER diagram for Task 4, branches, employees and joint accounts.

Reading the diagram

- A CUSTOMER OWNS many ACCOUNTs, and an ACCOUNT can have many owners. DateAdded is recorded for each owner.
- A BRANCH HOLDS many ACCOUNTs. Each ACCOUNT is held at one BRANCH.
- An EMPLOYEE WORKS_AT one BRANCH. A BRANCH has many EMPLOYEEs.
- One EMPLOYEE MANAGES one BRANCH.

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TASK 5 Moderate · University · About 25 minutes · Answers

Courses, lecturers and assignments

1. Entities

Entity	Reason
LECTURER	The university stores details about each lecturer.
COURSE	The university stores details about each course.
STUDENT	The university stores details about each student.
ASSIGNMENT	Each assignment has its own ID, title, due date and weight.

Not entities: Dr Ngata is one LECTURER, so she is not a separate entity. "Submission" is not an entity here either. It describes a student submitting an assignment, so it is shown as the SUBMITS relationship.

2. Attributes

Entity	Key attribute	Other attributes
LECTURER	StaffID	FirstName, LastName, Email
COURSE	CourseCode	Title, Credits
STUDENT	StudentID	FirstName, LastName, Email
ASSIGNMENT	AssignmentID	Title, DueDate, Weight

Grade, DateSubmitted and Mark are not listed here. Each one describes a student together with a course or an assignment, so they belong to relationships.

3. Relationships

Relationship	The two questions		Cardinality
LECTURER TEACHES COURSE	One LECTURER teaches how many COURSEs? One COURSE is taught by how many LECTURERs?	Many One	**1:N** Write 1 next to LECTURER and N next to COURSE.
STUDENT TAKES COURSE Relationship attributes: Grade	One STUDENT takes how many COURSEs? One COURSE has how many STUDENTs?	Many Many	**M:N** Write M next to STUDENT and N next to COURSE.
COURSE SETS ASSIGNMENT	One COURSE sets how many ASSIGNMENTs? One ASSIGNMENT belongs to how many COURSEs?	Many One	**1:N** Write 1 next to COURSE and N next to ASSIGNMENT.
STUDENT SUBMITS ASSIGNMENT Relationship attributes: DateSubmitted, Mark	One STUDENT submits how many ASSIGNMENTs? One ASSIGNMENT is submitted by how many STUDENTs?	Many Many	**M:N** Write M next to STUDENT and N next to ASSIGNMENT.

STUDENT TAKES COURSE: A student has a different grade in each course, and a course gives a grade to each student. So Grade belongs to TAKES.

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4. ER diagram

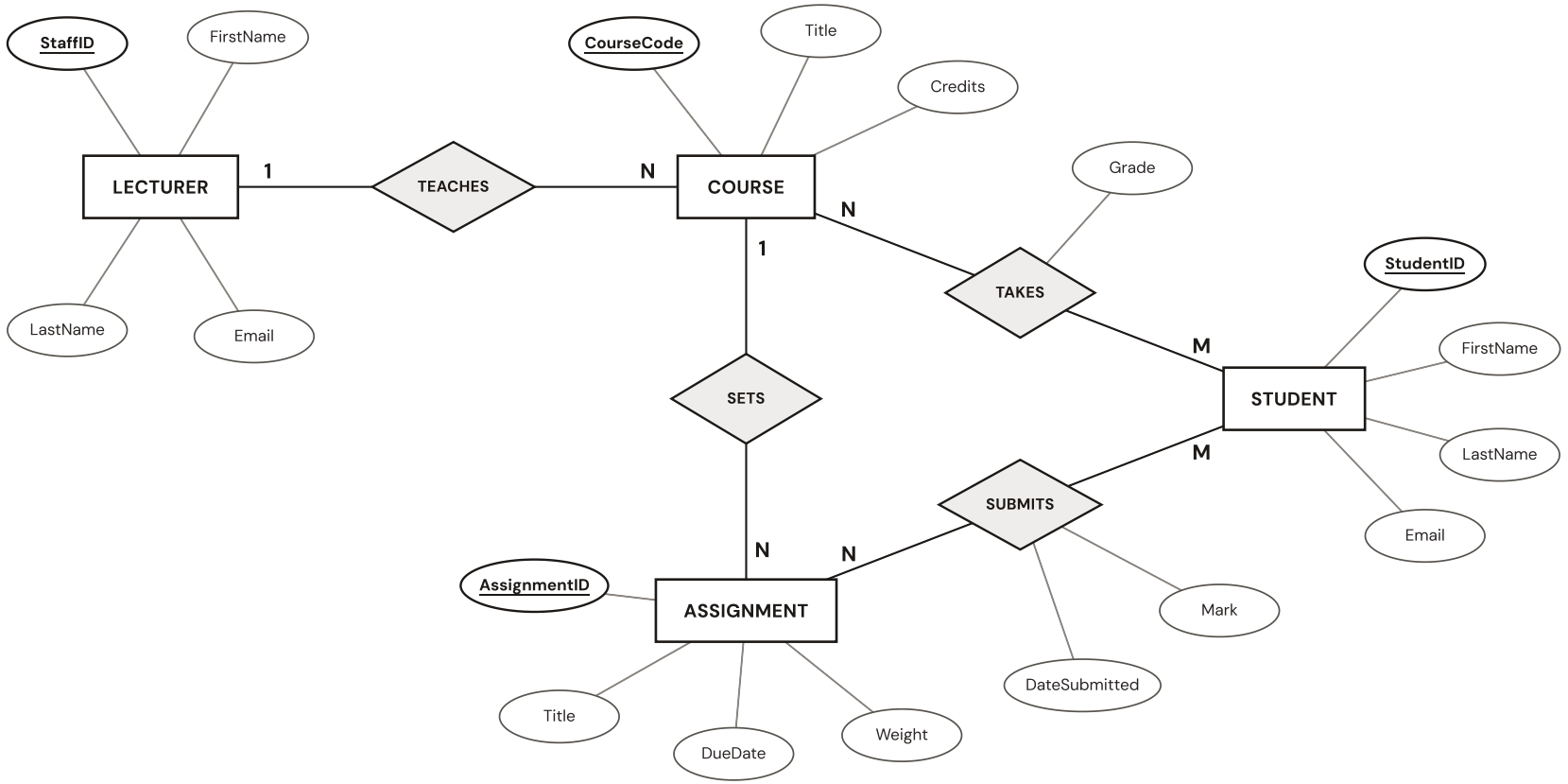


Figure 5. ER diagram for Task 5, courses, lecturers and assignments.

Reading the diagram

- A LECTURER TEACHES many COURSEs. Each COURSE has one LECTURER.
- A STUDENT TAKES many COURSEs. A COURSE has many STUDENTs. Each student gets a Grade for each course.
- A COURSE SETS many ASSIGNMENTs. Each ASSIGNMENT belongs to one COURSE.
- A STUDENT SUBMITS many ASSIGNMENTs. Each submission records a DateSubmitted and a Mark.

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TASK 6 Simple · Online shop · About 5 minutes · Answers

Customers and purchases

1. Entities

Entity	Reason
CUSTOMER	Hemi stores details about many customers.
PURCHASE	Hemi stores details about many purchases.

Not entities: The online shop and Hemi are not entities. Candles are not an entity here because the scenario stores nothing about them.

2. Attributes

Entity	Key attribute	Other attributes
CUSTOMER	CustomerID	Name
PURCHASE	PurchaseNumber	PurchaseDate

Two customers can have the same name, but not the same CustomerID.

3. Relationships

Relationship	The two questions		Cardinality
CUSTOMER MAKES PURCHASE	One CUSTOMER can make how many PURCHASEs?	Many	1:N
	One PURCHASE is made by how many CUSTOMERs?	One	Write 1 next to CUSTOMER and N next to PURCHASE.

Participation	The two questions	
CUSTOMER MAKES PURCHASE	Must every CUSTOMER have made a purchase?	**No, so CUSTOMER is partial (single line)**
	Must every PURCHASE be made by a customer?	**Yes, so PURCHASE is total (double line)**
	Draw a single line from CUSTOMER to MAKES and a double line from PURCHASE to MAKES.	

CUSTOMER MAKES PURCHASE: Cardinality and participation are different questions. Cardinality says a customer can make many purchases. Participation says a customer is allowed to have none.

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4. ER diagram

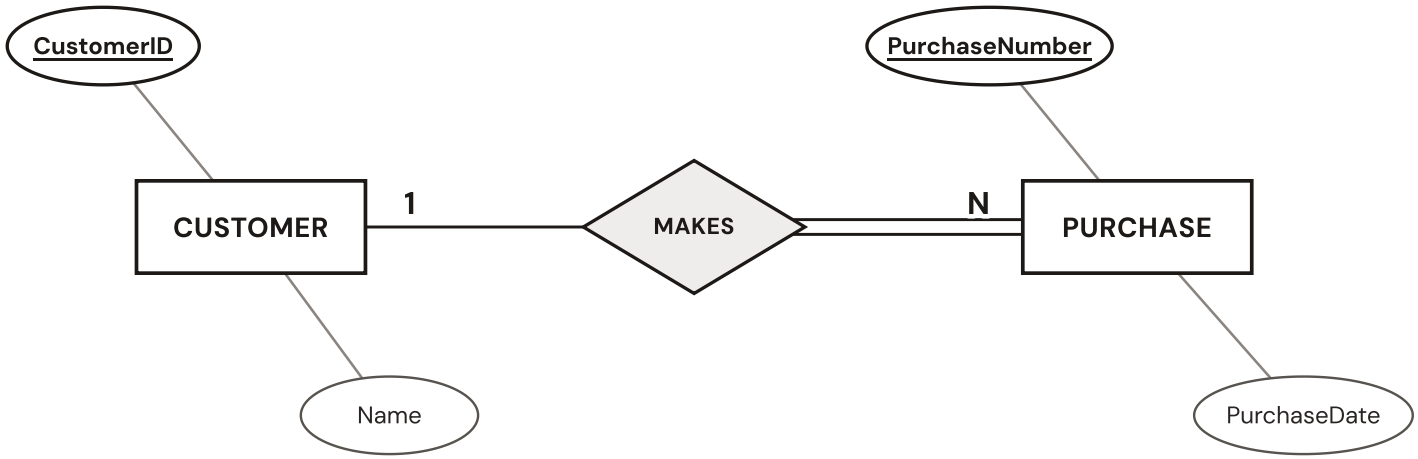


Figure 6. ER diagram for Task 6, customers and purchases.

Reading the diagram

- One CUSTOMER MAKES many PURCHASEs.
- Each PURCHASE is made by one CUSTOMER.
- A CUSTOMER may have no purchases (partial).
- Every PURCHASE must have a CUSTOMER (total).

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